

Yang Liu

✉ liuyang3489xyz@gmail.com ☎ 3463390270 📍 1885 EL Paseo St, APT 346, Houston-TX 77054

🌐 linkedin.com/in/yang-liu-278658195 🔄 https://yliu38.github.io/ (click here for personal page)

🏠 https://scholar.google.com/citations?user=opBShpUAAAAJ&hl=zh-CN&citsig=AKwN59pXyD7EPBUf2sYxvyK2S0g1

Education

- PhD in Biostatistics and Data Science**, *University of Texas Health Science Center at Houston* 01/2019 – 08/2025
Houston, United States
- Relevant Courses: Statistical Inference, Machine Learning, Survival Analysis, Clinical Trials
 - Honors: Caroline Ross Fellowship, Julius and Suzan Glickman Endowed Scholarship, Paul C. Boyle Research Award ×2, Dean's Excellence Scholarship, Schull Institute Summer Program Scholarship, James Turley Endowed Scholarship.
- Master in Cancer Biology**, *Seoul National University* 08/2015 – 08/2017
Seoul, South Korea
- Relevant Courses: Medical Statistics, Molecular Cell Biology
 - Honors: SNU Global Scholarship ×2
- Bachelor in Biotechnology**, *Heilongjiang Bayi Agricultural University* 08/2011 – 07/2015
Daqing, China
- Relevant Courses: Biochemistry, Molecular Biology
 - Honors: First-class Scholarship (1%), Second-class Scholarship (5%) ×2

Professional Experience

- Postdoctoral Researcher**, *AbbVie* 12/2025
Chicago, United States
- Multi-modal data integration for drug safety prediction through knowledge graphs
- Graduate Research Assistant**, *University of Texas MD Anderson Cancer Center* 01/2020 – 08/2025
Houston, United States
- **Engineered a novel deep learning framework** combining network-based feature engineering with **graph neural networks (GNNs)** to classify cancer phenotypes and identify biomarkers from gene expression data; secured \$519K Moon Shot funding (Briefings in Bioinformatics, 2025 📄)
 - **Developed a network-penalized statistical method** for biomarker identification in gene expression datasets with survival endpoints (bioRxiv, 2025 📄)
 - **Identified immune biomarkers** in a **phase II glioma trial** of pembrolizumab through gene expression analysis; enabled translational collaboration for treatment response assessment 📄
 - **Integrated scRNA-seq and scATAC-seq datasets** to elucidate functional consequences of germline risk variants in glioma progression; awarded \$461K R01 funding 📄
 - **Enhanced ICD-10 code prediction** from clinical notes by leveraging data-derived knowledge graphs and fine-tuned lightweight large language models (**LLMs**)
 - **Developed a RAG tool** to answer graduation-related queries at UTHealth for personalized student support (Huggingface spaces 📄)
 - **Built multimodal transformer models** for segmentation of Organ-at-Risk (OAR) regions using paired MRI and CT images (Github 📄)
- R&D Engineer**, *Berry Genomics* 11/2017 – 07/2018
Beijing, China
- Developed a BRCA diagnostic kit on NGS platform; optimized primer design for coverage
 - Coordinated cross-departmental collaboration for product development

Publications & Conferences

- **Authored** 5 first-author publications (4 published, 1 preprint) including 1 in *Briefings in Bioinformatics*, and 6 co-authored abstracts in *Neuro-Oncology* and *Molecular Cancer Therapeutics*
- **Presented posters** at JSM 2025, SNO Annual Meetings (2021, 2024), and Brain Tumor Center Retreat (2023)
- **Presented at** FY22 Brain Tumor Center Retreat, GraphConnect 2022, and Neo4j AWS GenAI Lab 2024
- **Selected papers**
- **Liu, Y.**, Huse, J., & Kannan, K. (2025). A network-based discovery of prognostic markers in recurrent IDH wild-type gliomas. *Frontiers in Genetics*, 16, Article 1672015. <https://doi.org/10.3389/fgene.2025.1672015> 📄
- **Liu, Y.**, Huse, J. T., & Kannan, K. (2025). Expression graph network framework for biomarker discovery. *Briefings in Bioinformatics*, 26(5), bbaf426. <https://doi.org/10.1093/bib/bbaf426> 📄
- **Liu, Y.**, Huse, J., & Kannan, K. (2025). Network regularized accelerated failure time models for robust biomarker identification. *bioRxiv*. <https://doi.org/10.1101/2025.05.30.657087> 📄
- **Liu, Y.**, Shin, D. Y., Oh, S., Kim, S., Koh, Y., & Kim, I. (2017). KML001 and doxercalciferol induce synergistic antileukemic effect in acute lymphoid leukemia cells. *Oncology Reports*, 38(1), 481-487. <https://doi.org/10.3892/or.2017.5652> 📄

Skills & Interests

- **Tools:** R, Python, SQL, Neo4j (Cypher), SAS, Tableau, Linux, HPC, Pytorch, CUDA, Version Control (Git)
- **Specializations:** Statistics, AI/ML, Bioinformatics, Bayesian, Simulation, Graph and Image Analysis, LLMs, Wet Lab
- **Languages:** Chinese (Native), English (Proficient), Korean (Proficient)
- **Interests:** Swimming, Hiking, Piano